Price Discrimination

In the note *The Economics of Product Variety,*[1] we examined the economic trade-offs affecting a firm's decision to offer one or more versions of a product to a segmented market. The underlying assumption in the note was that the various product versions are differentiated by their physical, functional, or performance characteristics. A product can also be differentiated by attributes such as place and/or time of purchase, duration between purchase and consumption, quantity purchased, and restrictions on consumption. Whether or not these attributes are included in the formal definition of the product, one thing is certain: they result in a different type of product "variety" and constitute a basis for *price discrimination* (where consumers who purchase and consume the product under different circumstances pay different prices).

This note provides a classification scheme for some of the more common examples of price discrimination. For each case that is discussed, the note also characterizes the appropriate market segments, the alternative product "versions," and the differences in the firm's economics from one version to another. A complete analysis of the optimal degree of price discrimination is, however, not presented. For that, the reader is referred to the note *The Economics of Product Variety*, the methodology in which applies—with some adaptation—to most cases in this note.

Motivation

Suppose you need a pint of milk and a bag of cookies (they are good for you). You have two choices: you can get into your car, drive half a mile to a supermarket, spend five minutes looking for the stuff and fifteen minutes in the "express" checkout lane, drive back half a mile, park your car, and—finally—eat the cookies and drink the milk; alternatively, you can hop across to the "mom-and-pop" shop around the corner, buy the milk and cookies in a jiffy, and dash back home. The choice would be obvious, were it not for the fact that the milk and cookies (yes, the same brands) are cheaper in the supermarket.

Why are the milk and cookies cheaper in the supermarket? Take first the cost-related explanations. One such explanation might be that the supermarket gets a quantity discount from suppliers and, therefore, has lower unit variable costs than the mom-and-pop store. Another explanation might be that the supermarket has more efficient operations and, therefore, lower

[1]*The Economics of Product Variety*, Harvard Business School Note #9-191-099 (Boson, Massachusetts: Harvard Business School, 1990)

Professor Anirudh Dhebar wrote this note as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

spoilage, inventory-carrying costs, overheads, and so on. The supermarket might also be located on cheaper land.

There is another explanation, and it has nothing to do with costs: From the customer's point of view, the mom-and-pop store is close by and convenient to get to—a convenience for which some people will pay a price premium. The mom-and-pop store maximizes contribution by extracting that premium from the convenience-loving consumers. Of course, the level of premium that the store can extract depends critically on the nature of competition. Indeed, competition plays a major role in the determination of prices at both the supermarket and the mom-and-pop store.

Returning to the purchase-convenience theme, let us say you value the convenience and are willing to pay the mom-and-pop store the price premium. Your next-door neighbor—who is retired and not as well off as you—thinks you are crazy. He would rather go to the supermarket. Your friend, who lives two doors away, agrees with your neighbor: she also would rather go to the supermarket. But not her neighbor, who thinks it is not worth his while going all the way to the supermarket just to save sixty cents (the price difference). And on and on and on. Some of your neighbors favor the supermarket; others, the mom-and-pop store.

Now suppose the mom-and-pop store and the supermarket are owned by the same firm. In that case, the common owner offers the same products, but at different prices to the consumers (one price in the supermarket, another price in the mom-and-pop store), and lets consumers *self-select* the purchase location on the basis of their valuation of the convenience benefits relative to the price differential (individual consumers have a willingness to pay for a product with convenient purchase and a different willingness to pay for the same product, but without convenient purchase; each consumer compares the two willingness-to-pay's with the respective purchase prices, and chooses the option that maximizes his or her consumer surplus). Since consumers are heterogeneous in their valuation of purchase convenience, some, like you, will buy the product from the mom-and-pop shop; others, like your neighbor, will make the trip to the supermarket.

Whether or not the common owner should offer the product in both the supermarket and the mom-and-pop store, and the prices at which it should offer the product in the two outlets depend on the degree of heterogeneity among the consumers with respect to their valuation of purchase convenience, the economics of establishing and managing multiple sales outlets, the costs of offering the product in each outlet,[2] the firm's ability to set viable and non-confusing prices for each product at each outlet, the economics and effectiveness of reaching out to different segments of the market, and the nature of competition.

In each of the following cases, you should identify the different product "versions" and market segments, and think about the economic considerations affecting the optimal degree of price discrimination. A cautionary note: for the discrimination to be effective, there must be a physical, practical, or legal impediment to the buyer of a cheaper version reselling to someone who would have bought a more expensive version. For example, it should not be worth your neighbor's while buying the cheaper milk and cookies from the supermarket and reselling them to you.

"Charge People What They Can Afford to Pay"

Your friend is a dentist in a small town in the midwest. You visit him during Christmas break and tell him about all the wonderful things you have learned at business school. Somehow, pricing is one thing that is mentioned. After listening to you for a while, he says: "Let me tell you a

[2]Price setting here can be as complicated as price setting in the case of product variety. For a detailed discussion of product variety, price setting, and consumer self-selection, please refer to the note *The Economics of Product Variety*.

thing or two about pricing. I usually charge $30 for a tooth extraction, but sometimes I give people a break. The other day, I had someone in the clinic who was homeless. I didn't charge her a cent."

"But I am no saint," he continues. "Just last week, someone pulled up in a Jaguar—someone from out of town. She was passing through and suddenly had tooth trouble. I had to extract her tooth. When it was time to bill her, I looked at her, her clothes, her car, told my assistant '$50', and went on to the next patient. She didn't complain; I didn't feel guilty."

Your friend is price discriminating.

Spatial Discrimination

You are a London resident, off on a holiday to the Far East. Your first stop: Singapore. Before leaving London, you bought yourself a camera to take pictures of the trip. (An auto-everything affair, frightfully expensive when first introduced, but now much cheaper.) Camera in hand, you are checking out the sights of Singapore. You find yourself in front of a camera store's display window, and are shocked: there is a camera there, just like yours (the model number is different, but you know firms do that all the time—call the same product different things in different markets), and—you pull out a calculator, convert currencies, and adjust for taxes—considerably cheaper than your camera. You feel cheated.

We have global markets, global brands, and global products, but not global prices. Prices for the same product may vary from country to country, region to region, even neighborhood to neighborhood. There are several supply- and business environment-related reasons for the differences in prices (e.g., fluctuations in exchange rates and differences in tax rates, transportation costs, rents, quantity discounts, overhead costs, and competition). There is a demand-related reason as well: consumers in different geographical markets are different, with different preferences, needs, incomes, budgets, and willingness-to-pay. These demand differences provide an important basis for spatial price discrimination—the charging of different prices on the basis of location.

Bundling

You want lunch and walk across to the local "Soup, Salad, Etc." restaurant. What you want is a hearty salad. You look at the menu: a salad is $4.99 (all you can eat, but please do not share!). Someone comes to take your order, and—while doing so—recommends the soup of the day. It is winter and you are freezing, but you do not want to spend an extra $1.95 (12-oz.; $2.50, 16-oz.) for a bowl of hot soup. Not to worry: you can get the salad-and-soup for only $5.49. You are delighted: you were willing to spend $4.99 for a salad; now, for an additional 50¢, you can get the $1.95-soup as well.

You dragged your friend along for lunch. She is not very hungry; a bowl of soup is all she wants. She looks at the price list: $1.95 for a 12-oz. bowl, $2.50 for a 16-oz. bowl. Clearly, the $2.50 bowl is better value for the money. She listens to the "salad-and-soup for only $5.49" sales pitch and thinks: "I haven't had a salad for a long time, and I can use the vitamins … In any case, I'll get a $5-salad for an extra three dollars." She, too, is delighted.

You wanted a salad and were prepared to spend $4.99. Your friend wanted a soup and was prepared to spend $1.95 (or $2.50). Each ended up spending $5.49 for salad-and-soup.

"Soup, Salad, Etc." is doing quite well, thank you: whereas you and your friend intended to spend—in round numbers—$7 (or $7.50), you ended up spending $11. True, the extra soup (for you) and the extra salad (for your friend) added to the restaurant's cost, but not $4 more. True, you got to drink soup and your friend to eat a salad, but neither of you was that keen on the extra food.

What we have here is another example of price discrimination: by bundling the salad and the soup, the restaurant got you and your friend to spend more than you would otherwise have, and—as long as the resulting increase in revenues is greater than the increase in costs—thereby increased its contribution.

Examples of bundling abound: series tickets for concerts, sports, the theater; suits vs. a pair of trousers and a jacket; air fare-plus-hotel-plus-theater travel packages; "fully- loaded" cars; "fully-applianced" kitchens; "special deals" for a camera-with-lenses or a computer-with-hard disk-with-monitor-with-popular-software; and so on.

Price Skimming

You must—absolutely, positively must—buy every new hi-tech gizmo the day its hits the market. Take the new digital audio tape (DAT) recorder. You went out and bought yourself a DAT machine during the first week of its introduction—for $1,199. You know you should not have done this: last time, you bought a compact-disc player—also in week one—for $695; your friend bought a fancier model, albeit three years later, for $295; and you decided "Never again." But there you go again!

For almost every new product, some of us must buy the product the moment it is introduced, while others prefer to wait for six months, a year, several years—until the hype goes away, product designs mature, "bugs" are cleaned out, and prices drop.

Prices drop for many reasons. One, as the product ages and the cumulative production volume builds up, the manufacturer's production costs tend to decline (for example, because of "experience-curve" effects). Two, competitive pressures are usually more significant in the later stages of a product's life. Three, firms know some of us will wait before buying a new product, while others will not. And four, firms also know that some of us will pay a premium for the immediate consumption of a new product—perhaps because we have a greater and more immediate need for the product, perhaps because we can afford it, or perhaps because we enjoy the aura of exclusivity. Be as it may, what we observe is price discrimination over time, with the early buyers paying more and the later buyers less. This form of price discrimination is often referred to as "price skimming."

Peak-Load Pricing

Your friend works in the pricing department at Acropolis Edison, the primary supplier of electricity in the Acropolis metropolitan area in the southern United States. When she hears you are studying price discrimination, she says: "At Acropolis Edison, we do that all the time. We have three types of customers: residential, commercial, and industrial. All of them make our life difficult in the summer: during daylight hours, industries turn on their equipment; commercial customers, their air conditioners; and residential customers, their air conditioners. On really hot days, the peak energy demand is well above our generation capacity, and we have to buy energy from the national grid. The purchased energy is more expensive than our own energy.

"Fortunately, Acropolis's consumers are heterogeneous in their willingness to pay for energy—in terms of quantity consumed and the timing of consumption. Thus, industrial establishments are able to adjust equipment usage so that their energy needs peak at nighttime; commercial consumers are relatively inelastic with respect to the price of energy and the timing of their energy consumption; and residential consumers are somewhat elastic with respect to the timing and the amount of energy consumption. Accordingly, we have different rates for residential vs. nonresidential consumers and for day vs. night consumption. In particular, our prices are higher for nonresidential consumers and for daytime consumption than for nighttime consumption."

By charging more for peak demand, Acropolis Edison is making its consumers self select the timing of their consumption so that it is optimal for them and at a more convenient time for the supplier. Acropolis Edison is also price discriminating.

Air Fares, Hotel Fares, Etc.

"Hotels Gouge Graduating Students" screamed a headline in a recent issue of a business-school student newspaper.[3] The article began:

> Capitalizing on strong demand for rooms during Harvard's commencement, local hotels have increased prices to record levels for the event. A survey of local establishments uncovered significant rate increases, required minimum stays, and other hidden changes.

A caption in the article read: "The [name of hotel] currently has 160 people on its waiting list, all willing to pay … $840 for what could normally be had for $175."

Let us consider a different setting. Two brothers, Orville and Wilbur, buy tickets to fly Boston-San Francisco-Boston. Orville needs to get to San Francisco next week, hates stopovers, cannot leave before Tuesday, and must return Friday night. Wilbur, on the other hand, can go anytime in the next six months, is not particular about the route or the date or time of departure, and will stay over a weekend. Orville's ticket costs $1,474; Wilbur's, $495.[4]

One reason why the hotel cited in the newspaper article charged a higher rate during Harvard's commencement and why Orville's ticket was more expensive is that the demand curves for a hotel room during commencement or an aircraft seat on a Friday are different from the demand curves for a room during non-commencement or an aircraft seat on, say, Saturday afternoon. Specifically, the demand is higher during peak times—higher in terms of number of consumers and their willingness-to-pay. The high prices merely reflect the surge in demand.

There is a second reason for the price differences: in both examples, the firm supplying the product (a room in a hotel, a seat on an aircraft, *et cetera*) offered a price break to those consumers who were willing to commit purchase at an early date and who were willing to put up with some inconveniences or restrictions on product consumption. Unlike the other examples of price discrimination described in this note, it is difficult to make a case for price differences in these examples on the basis of cost differences (the cost of flying a business person in a suit is no different from the cost of flying a leisure traveler in a pair of shorts). The prices are different largely because some of us have a greater need for a hotel room or for air travel, can afford to pay more, cannot plan far ahead, have tighter constraints on the travel schedule, and so on.

Quantity Discounts and Multi-Part Tariffs

Example 1 You have just written a report and need to have it copied and bound. You go to the neighborhood copy store to check on prices. It will cost you $9.95 if you want one copy and $6.95 a copy if you want 100 or more copies.

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[3]K. Kharbanda. "Hotels Gouge Graduating Students," *The Harbus News,* Harvard Business School, Boston, Massachusetts, September 24, 1990.

[4]For a detailed description of the airline pricing and yield management problem, please refer to the case *American Airlines, Inc.: Revenue Management,* Harvard Business School Case #9-190-029 (Boston, Massachusetts: Harvard Business School, 1990)

Example 2 You have just moved into a new town and are inquiring about a phone connection. You have two choices: (1) $4.95 per month, with 30 free "message units" and 10¢ for every additional message unit; or (2) $14.95 per month, with an unlimited number of free local calls.

Example 3 There is a new video store in your neighborhood. It has the following price schedule: you can either rent a videotape at a time for $2.75 a night, or you can get a $25-annual membership with twelve free movies (one a month) and $2 for every additional video rental.

Example 4 It is winter and there is plenty of snow on the ground. You decide to go skiing. The slope you go to offers two alternatives: (1) $30-a-day, with unlimited ski lift rides and use of slopes; or (2) a $5-entry charge and $5 for every ride.

In all these cases, the supplier is price discriminating on the basis of quantity consumed. Typically, the supplier faces two types of costs: costs related to the number of customers serviced, and costs related to the volume of the product consumed. The larger the consumption volume, the lower the supplier's unit cost (for example, because there are fewer setups). So, cost difference is one justification for the price discrimination. A second justification might be that the quantity discounts are necessary for the firm to retain the custom of large-volume consumers—consumers who might otherwise go to the firm's competitors, leaving it without a volume base from which to derive, say, scale economies. There is a third justification: consumers differ in the volume of the product they wish to consume and in their willingness to pay for each unit of consumption, and the supplier uses this information when developing a pricing scheme.